

Amadou Siaka SANGARE

amadousangare21@gmail.com | linkedin.com/in/amadou-siaka-sangare | amadoussangare.com

EDUCATION

- **Université Paris-Saclay** Orsay, France
Doctoral program
2025 - 2028
Topic: Controllable image and video generation
Advisors: Adrien Maglo, Mohamed Chaouch, Bertrand Luvison
- **Télécom Paris** Palaiseau, France
Master of Science in Engineering
2021 - 2025
One of France's top prestigious general engineering schools in the Grandes Ecoles system. Equivalent to a top-tier university specializing in engineering studies
Courses: Signal processing, time series, statistics, machine learning, deep learning, reinforcement learning, optimization, advanced algorithms, games theory, economy
GPA: 3.93
- **Université Paris-Saclay** Orsay, France
Licence Double Diplômes Informatique, Mathématiques
2020 - 2021
Dual degree in computer science and mathematics, third year of bachelor's degree, intensive bi-disciplinary program
Courses: Differential calculus, probability, statistical inference, compilation, machine learning, numerical modeling and simulations with Python, matrix algebra with computation
GPA: 4.0
- **Université Grenoble Alpes** Valence, France
Licence Mathématiques et Informatique
2018 - 2020
First and second year of a bachelor's degree in mathematics and computer science.
Courses: Linear algebra, calculus, topology, probability, algorithms, object-oriented programming, functional programming
GPA: 4.0

ACHIEVEMENTS & HONORS

- **Oral presentation of a paper to the Junior Conference on Data Science and Engineering** 2024
- **Winner of Mali's excellence scholarship to study in France** 2018

SKILLS SUMMARY

- **Programming:** Java, Python, C, C++, JavaScript
- **Libraries/Frameworks:** Scikit Learn, PyTorch, TensorFlow, Hugging Face, NodeJS, React.js
- **Tools:** Git, Docker, Kubernetes, SLURM

PROFESSIONAL EXPERIENCE

- **PhD Candidate at CEA List** Palaiseau, France
Topic: Controllable image and video generation
Advisors: Adrien Maglo, Mohamed Chaouch, Bertrand Luvison
April 2025 - April 2028
- **Research Intern in Computer Vision at University of Liège** Liège, Belgium
Subject: Leveraging motion for self-supervised object detection or classification
Advisors: Anthony Cioppa, Silvio Giancola and Marc Van Droogenbroeck
November 2024 - February 2025
Working on a research project where the goal is to develop a self-supervised learning framework in order to learn features that capture object motion in videos. The final goal is to transfer these representations to propagation downstream tasks such as video object segmentation, human pose propagation and semantic part propagation.
 - **Responsibilities:** Fully in charge of the research project and having many responsibilities including doing literature review, coding, running experiments on GPU clusters and presenting findings.
- **Research Intern at CEA List** Palaiseau, France
Subject: Automatic image generation for team sports matches
Advisors: Adrien Maglo and Mohamed Chaouch
April 2024 - October 2024
Worked on a research project where the goal is to build an image generative model for generating realistic team sport matches (e.g soccer games). The main focus is to control the image generation by adding the ability to control many aspects of the scene including player positions and poses, camera calibration, etc.
 - **Responsibilities:** Fully in charge of the research project and having many responsibilities including doing literature review, coding, running experiments on GPU clusters and presenting findings.
- **AI research project at Télécom Paris** Palaiseau, France
Subject: Contrastive Learning in Graph Neural Networks with Optimal Transport
Advisors: Jhony H. Giraldo and Fragkiskos D. Malliaros
September 2023 - March 2024
Worked on a subgraph-based contrastive learning framework for graph self-supervised learning using Fused Gromow-Wasserstein distance at Télécom Paris' LTCI lab.

- **Coding:** In charge of implementing the theory and running experiments with GPUs
- **Research review:** In charge of keeping track of potential research papers related to our research in order to find interesting insights and approaches.
- **Achievement:** Submitted a paper to a machine learning journal.
- **Laboratoire d'Informatique de Grenoble (LIG) Internship** Grenoble, France
May 2019 - July 2019
 - **Subject:** *Automatic Automaton Learning Algorithms*
 - **Optimization of learning algorithms:** Optimized hyper-parameters of different algorithms
 - **Determination of the best algorithm:** Compared optimized algorithms on benchmarks

PUBLICATIONS

- **Improving Controllable Generation: Faster Training and Better Performance via x_0 -Supervision:** CVPR 2026
Authors: **Amadou S. Sangare**, Adrien Maglo, Mohamed Chaouch, Bertrand Luvison
Keywords: Controllable generation, Diffusion Models, Flow Matching
- **Towards fine-grained spatial control for soccer game image generation:** CVPRW 2025
Authors: **Amadou S. Sangare**, Adrien Maglo, Baptiste Engel, Mohamed Chaouch
Keywords: Image generation, diffusion models, soccer
- **A Fused Gromov-Wasserstein Approach to Subgraph Contrastive Learning:** TMLR 2025
Authors: **Amadou S. Sangare**, Nicolas Dunou, Jhony H. Giraldo, Fragkiskos D. Malliaros
Keywords: Self-supervised learning, contrastive learning, graph neural networks, optimal transport

PROJECTS

- **GAN-based super resolution:** Implementation of the research paper of (Ledig et al., 2017) that introduces a GAN-based super resolution model. The goal was to understand Generative Adversarial Networks. Tech: Python, PyTorch. Link: <https://github.com/sangaram/gan>
- **Image generation with VAE's:** Implementation of the paper of (Kingma et al., 2014) that introduces a new technique for generative modeling based on a latent variable model and a variational approximation of the true prior distribution with a deep neural network. Tech: Python, PyTorch. Link: <https://github.com/sangaram/vae>
- **Diffusion model:** Implementation of an unconditional Denoising Diffusion Probabilistic Model (DDPM) introduced by (Ho et al., 2020) with a UNet backbone and application to the CelebA-HQ-32 dataset. Tech: Python, PyTorch. Link: <https://github.com/sangaram/diffusion>
- **Speech to Speech Machine Translation:** Development of a speech to speech machine translation model with a cascaded approach consisting in performing speech to text followed by machine translation and text to speech. Tech: Python. Link: <https://github.com/sangaram/S2SMT>

LANGUAGE PROFICIENCY

- **Bambara:** Native
- **French:** Fluent
- **English:** Fluent, Linguaskill (score: 180+), TOEFL-iBT (score: 90/120)
- **Arabic:** Notions